

Assemble Your Toolbox

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1 Purpose

This document was created for the **Assemble Your Toolbox** course project in the Johns Hopkins University **Data Scientist's Toolbox** course on Coursera.

The purpose of this project is to demonstrate that the basic tools used in the Data Science Specialization have been installed, configured, and connected successfully.

The tools demonstrated in this project are:

- R
- RStudio
- Git
- GitHub
- R Markdown
- CRAN packages
- Bioconductor packages

2 R and RStudio Setup

R and RStudio were installed locally and used to create this project. The project was managed through an RStudio Project folder and connected to GitHub using Git version control.

The local R version and working directory are shown below.

```
R.version.string
```

```
## [1] "R version 4.5.2 (2025-10-31 ucrt)"
```

```
getwd()
```

```
## [1] "C:/Users/travi/Documents/R-Programming-Course"
```

3 Package Setup

The course introduced package installation from CRAN, Bioconductor, and GitHub. The packages below were installed or used as part of the course setup process.

```
course_packages <- c(
  "ggplot2",
  "devtools",
  "lme4",
  "BiocManager",
  "GenomicFeatures",
  "rmarkdown",
  "knitr"
)

package_check <- data.frame(
  Package = course_packages,
  Installed = course_packages %in% rownames(installed.packages())
)

package_check
```

```
##           Package Installed
## 1      ggplot2      TRUE
## 2     devtools      TRUE
## 3         lme4      TRUE
## 4    BiocManager      TRUE
## 5 GenomicFeatures      TRUE
## 6      rmarkdown      TRUE
## 7         knitr      TRUE
```

The main setup packages were also loaded to confirm that the local R environment is working.

```
library(ggplot2)
library(lme4)
library(BiocManager)
library(GenomicFeatures)

cat("Course setup packages loaded successfully.")
```

```
## Course setup packages loaded successfully.
```

4 R Markdown Demonstration

R Markdown combines regular written explanation, R code, and code output in one reproducible document. This file was created in RStudio and knitted into a rendered output document.

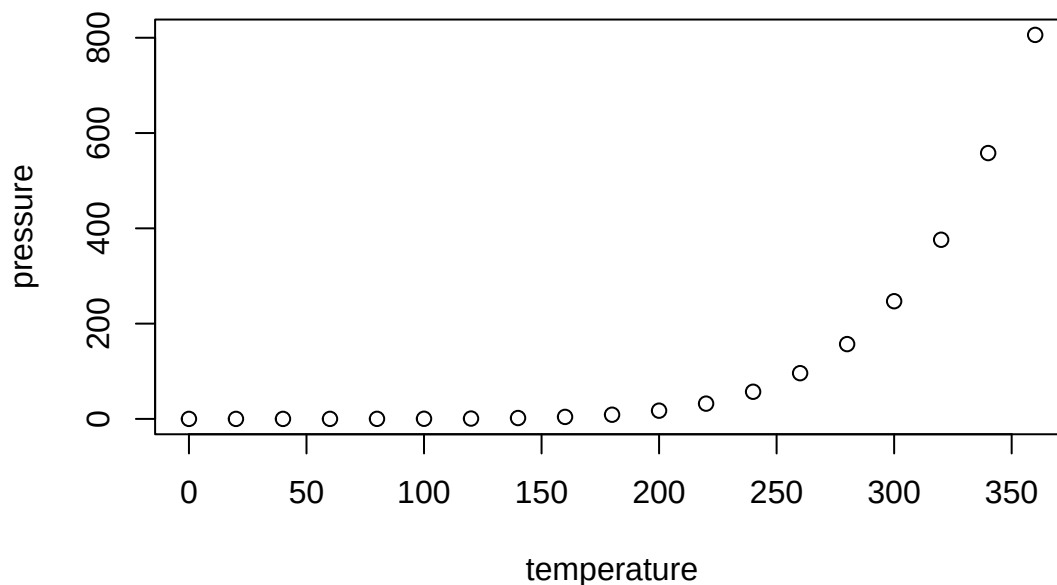
The simple R code below confirms that code chunks run inside the R Markdown document.

```
x <- c(1, 2, 3, 4, 5)
mean(x)
```

```
## [1] 3
```

The plot below confirms that R Markdown can also include R-generated visual output.

```
plot(pressure)
```



5 Git and GitHub Setup

Git was used for local version control, and GitHub was used as the remote repository.

The workflow demonstrated in this project was:

1. Create and edit files in RStudio.
2. Stage files using Git.
3. Commit changes locally with a descriptive message.
4. Push committed changes to GitHub.
5. Confirm that the files appear in the GitHub repository.

The GitHub repository for this project is:

<https://github.com/treinart/JHU-Data-Scientists-Toolbox>

6 Setup Evidence

The screenshots below document the setup and version control workflow. The screenshots are stored in the `images/` folder of the repository and are included here as supporting evidence.

6.1 RStudio Git Staging

This screenshot shows the RStudio Git pane with project files prepared for commit.

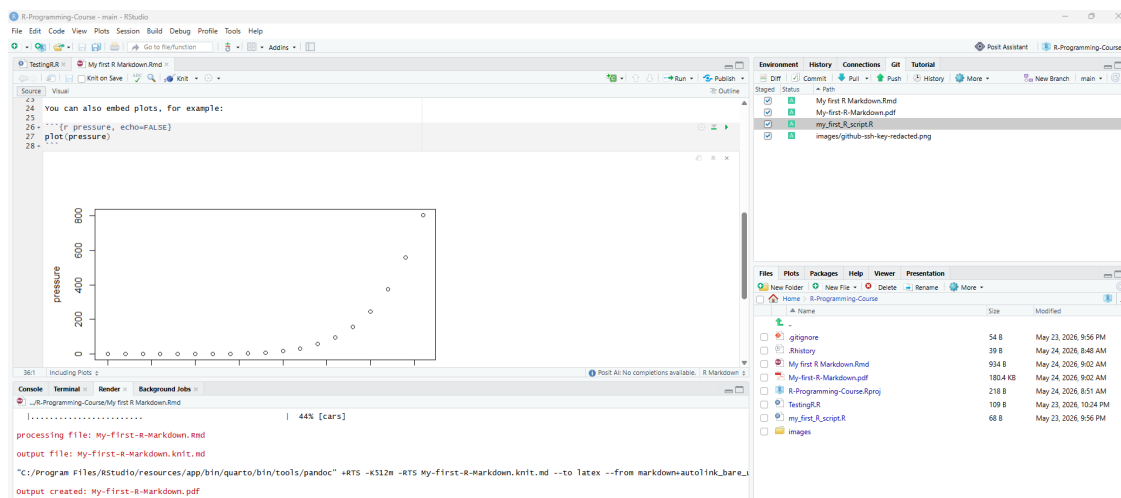


Figure 1: RStudio Git pane showing staged project files.

6.2 RStudio Review Changes and Commit Message

This screenshot shows the RStudio review window and the commit message used for the project files.

6.3 Git Push Confirmation

This screenshot shows the Git push confirmation from RStudio.

6.4 GitHub SSH Key Setup

This screenshot shows the GitHub SSH key setup page with sensitive details redacted.

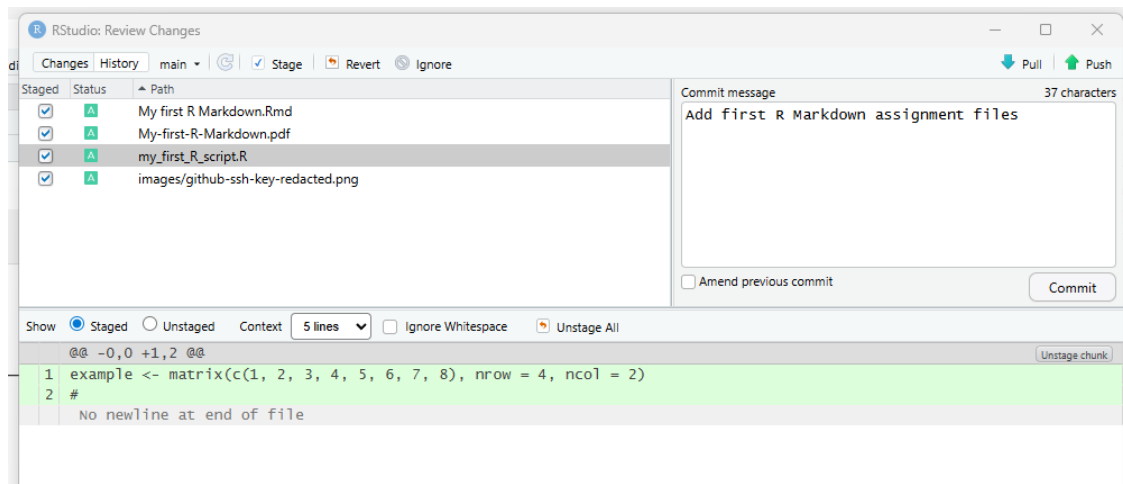


Figure 2: RStudio Review Changes window with commit message.

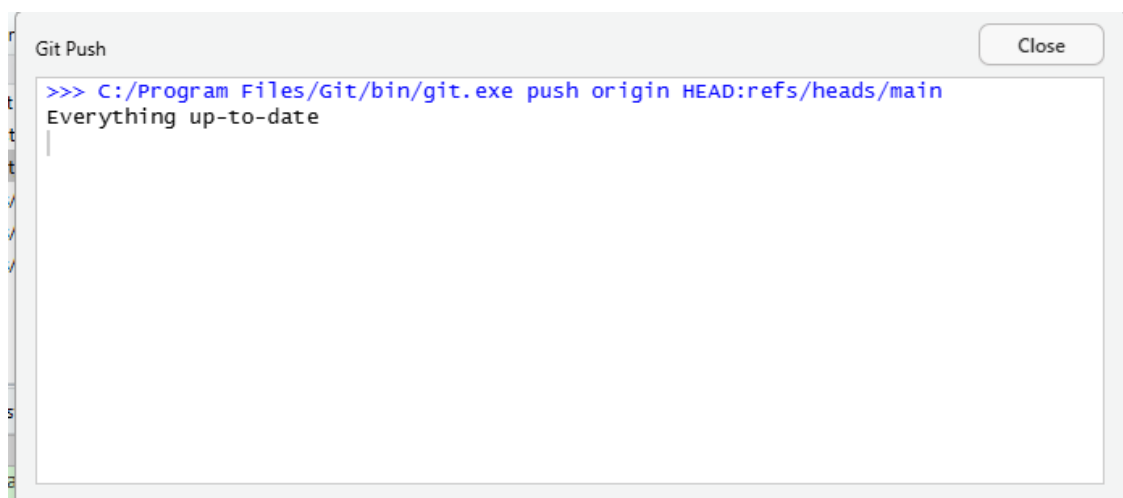


Figure 3: RStudio Git push confirmation.

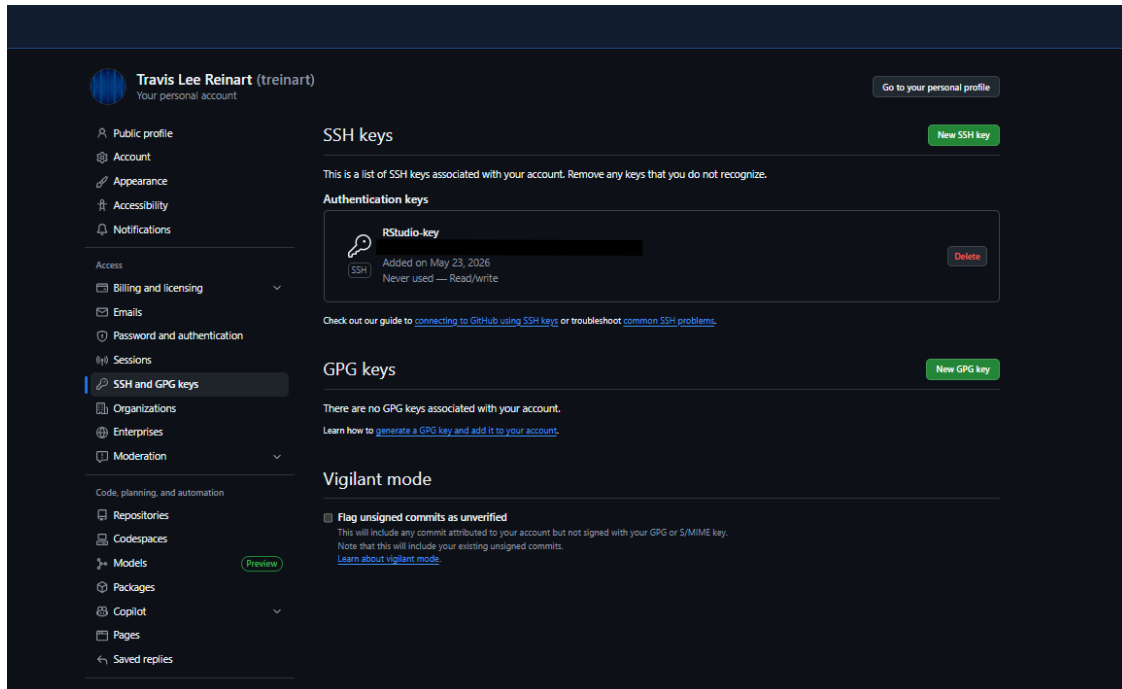


Figure 4: Redacted GitHub SSH key setup evidence.

6.5 GitHub Repository After Push

This screenshot shows the GitHub repository after the project files were pushed.

Image not found: 'images/github-repo-files-after-push.png'

7 Conclusion

This project demonstrates that the required course toolbox has been assembled. R and RStudio are installed locally, R packages can be installed and loaded, R Markdown can be created and rendered, Git is tracking the project files, and GitHub is receiving pushed commits from the local RStudio project.

The repository provides a working setup foundation for the rest of the Johns Hopkins Data Science Specialization.